
Report 5

Breast-cancer-prediction

Dataset = WISCONSIN DIAGNOSIS BREAST CANCER(WDBC)

Friday, 27th july 2024

Content:

Sorting p-values and calculating their effect sizes base of the cohen'd equation.

features	p_values	0>0.5 0<0.5	t_test	cohen_d
concave points_worst	1.969100e-124	We Reject the Null Hypothesis: There is a significant difference in means.	31.054555	2.692647
perimeter_worst	5.771397e-119	We Reject the Null Hypothesis: There is a significant difference in means.	29.965717	2.598237
concave points_mean	7.101150e-116	We Reject the Null Hypothesis: There is a significant difference in means.	29.354319	2.545225
radius_worst	8.482292e-116	We Reject the Null Hypothesis: There is a significant difference in means.	29.339082	2.543903
perimeter_mean	8.436251e-101	We Reject the Null Hypothesis: There is a significant difference in means.	26.405213	2.289516
area_worst	2.828848e-97	We Reject the Null Hypothesis: There is a significant difference in means.	25.721590	2.230242
radius_mean	8.465941e-96	We Reject the Null Hypothesis: There is a significant difference in means.	25.435822	2.205463
area_mean	4.734564e-88	We Reject the Null Hypothesis: There is a significant difference in means.	23.938687	2.075651
concavity_mean	9.966556e-84	We Reject the Null Hypothesis: There is a significant difference in means.	23.103963	2.003275

concavity_worst	2.464664e-72	We Reject the Null Hypothesis: There is a significant difference in means.	20.897175	1.811931
compactness_mean	3.938263e-56	We Reject the Null Hypothesis: There is a significant difference in means.	17.698392	1.534574
compactness_worst	7.069816e-55	We Reject the Null Hypothesis: There is a significant difference in means.	17.445374	1.512636
radius_se	9.738949e-50	We Reject the Null Hypothesis: There is a significant difference in means.	16.396351	1.421678
perimeter_se	1.651905e-47	We Reject the Null Hypothesis: There is a significant difference in means.	15.934158	1.381603
area_se texture_worst	5.895521e-46 1.078057e-30	We Reject the Null Hypothesis: There is a significant difference in means.	15.609343 12.230981	1.353439 1.060512
smoothness_worst	6.575144e-26	We Reject the Null Hypothesis: There is a significant difference in means.	11.066747	0.959564
symmetry_worst	2.951121e-25	We Reject the Null Hypothesis: There is a significant difference in means.	10.902304	0.945306
texture_mean	4.058636e-25	We Reject the Null Hypothesis: There is a significant difference in means.	10.867201	0.942262
concave points_se	3.072309e-24	We Reject the Null Hypothesis: There is a	10.642498	0.922779

		significant difference in means.		
smoothness_mean	1.051850e-18	We Reject the Null Hypothesis: There is a significant difference in means.	9.146099	0.793031
symmetry_mean	5.733384e-16	We Reject the Null Hypothesis: There is a significant difference in means.	8.338312	0.722990
fractal_dimension_worst	2.316432e-15	We Reject the Null Hypothesis: There is a significant difference in means.	8.151316	0.706776
compactness_se	9.975995e-13	We Reject the Null Hypothesis: There is a significant difference in means.	7.297077	0.632708
concavity_se	8.260176e-10	We Reject the Null Hypothesis: There is a significant difference in means.	6.246157	0.541585
fractal_dimension_se	6.307355e-02	We Accept the Null Hypothesis: There is no significant difference in means.	1.862330	0.161477
smoothness_se	1.102966e-01	We Accept the Null Hypothesis: There is no significant difference in means.	-1.599365	-0.138676
fractal_dimension_mean	7.599368e-01	We Accept the Null Hypothesis: There is no significant difference in means.	-0.305711	-0.026507
texture_se	8.433320e-01	We Accept the Null Hypothesis: There is no significant difference in means.	-0.197724	-0.017144
symmetry_se	8.766418e-01	We Accept the Null	-0.155298	-0.013465



		Hypothesis: There is no significant difference in means.		
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Results:

1. between **25** features that were **important** features of the t-test. Here is the importance rank of them:

1	concave points_worst	6	area_worst	11	compactness_me an
2	perimeter_worst	7	radius_mean	12	compactness_wor st
3	concave points_mean	8	area_mean	13	radius_se
4	radius_worst	9	concavity_mean	14	perimeter_se
15	area_se texture_worst	19	concave points_se	23	compactness_se
16	smoothness_wors t	20	smoothness_mea n	24	concavity_se
17	symmetry_worst	21	symmetry_mean		
18	texture_mean	22	fractal_dimension _worst		

2. **5** features were **not important** features of the t-test.

1	fractal_dimension_s e	4	texture_se
2	smoothness_se	5	symmetry_se
3	fractal_dimension_ mean		